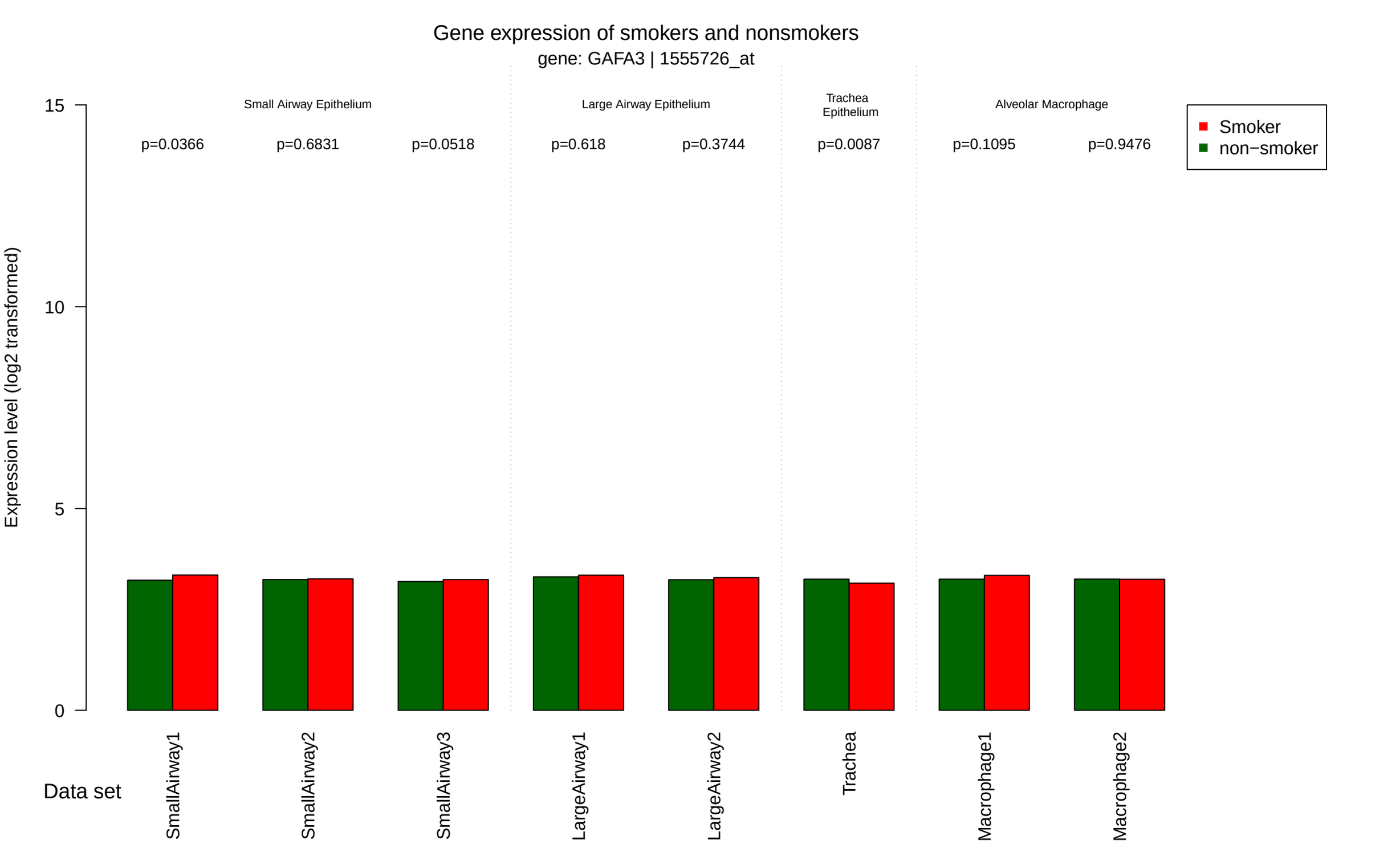
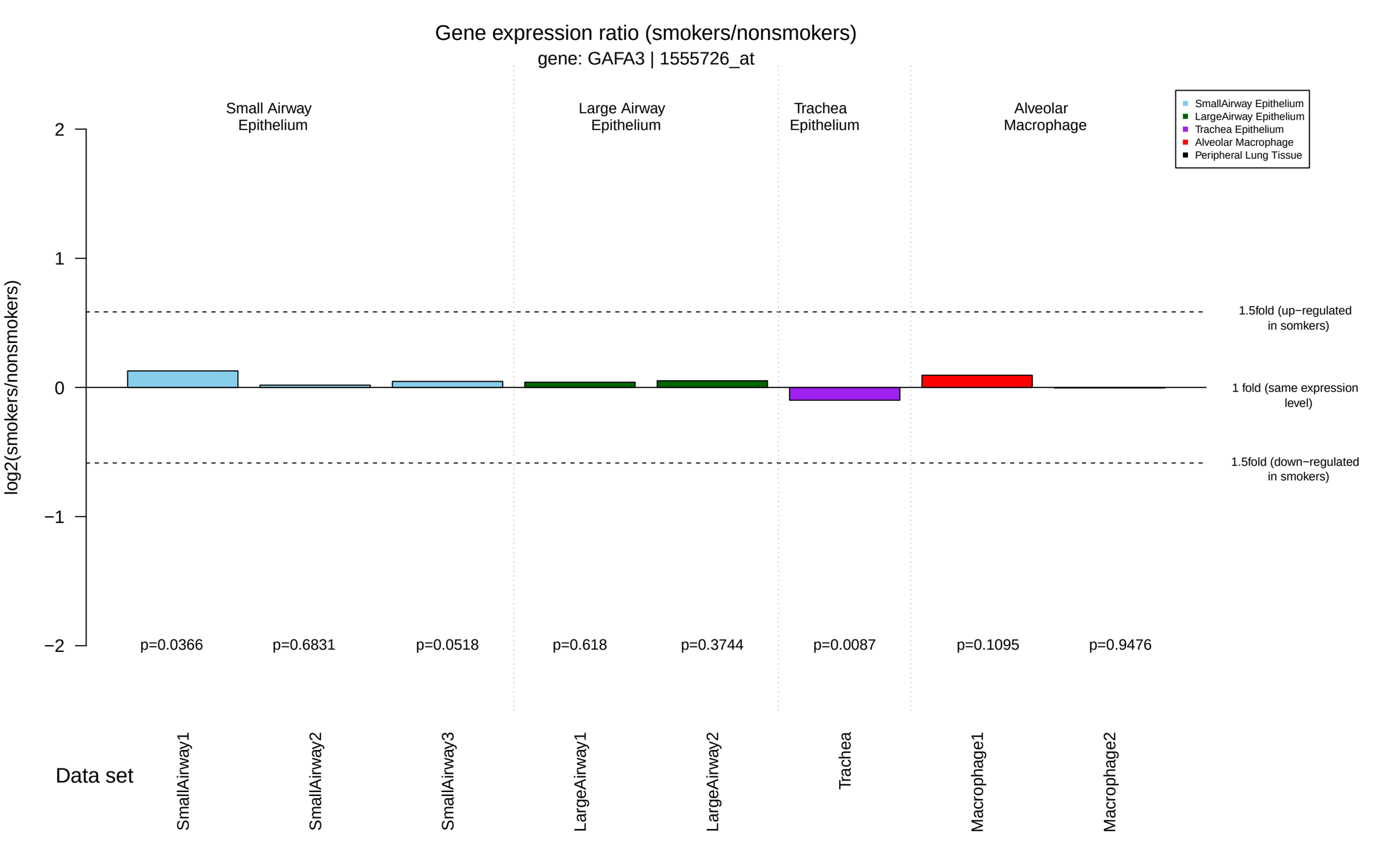
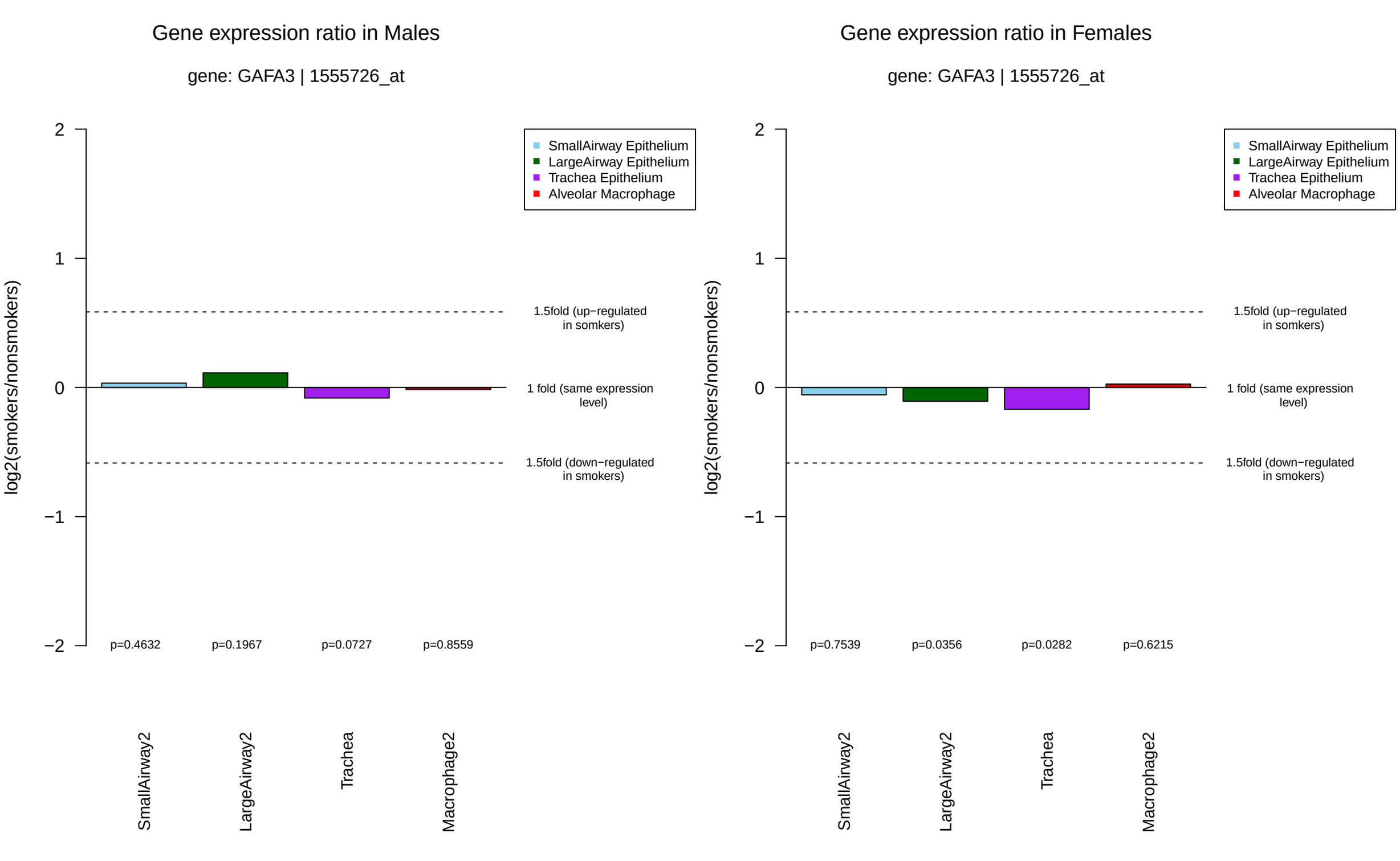
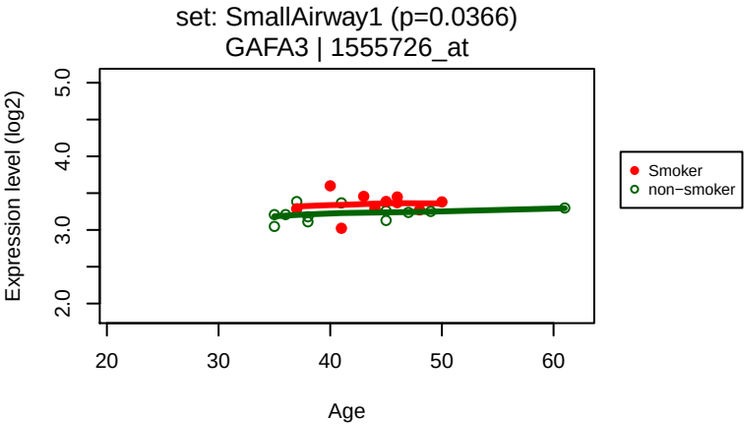
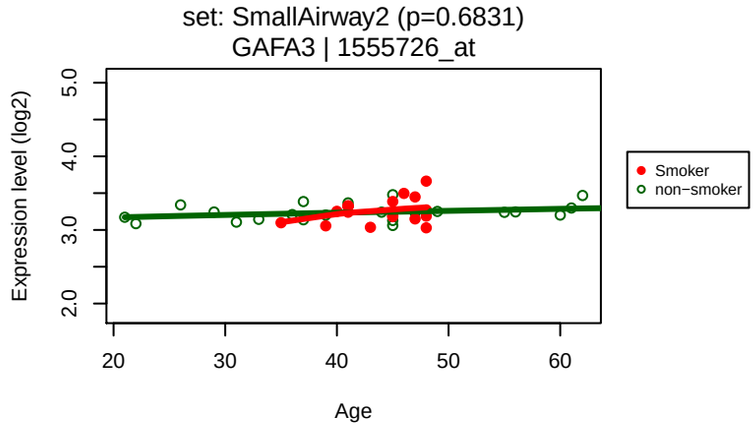
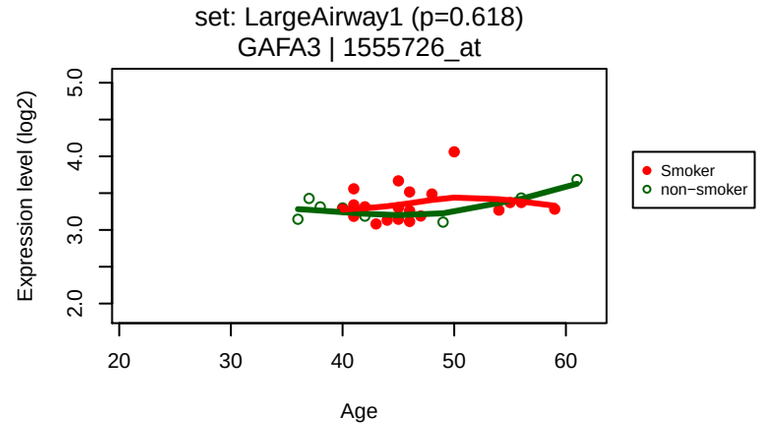

Gene expression of smokers and nonsmokers
gene: GAFA3 | 1555726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0366
p=0.6831
p=0.0518
p=0.618
p=0.3744
p=0.0087
p=0.1095
p=0.9476
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

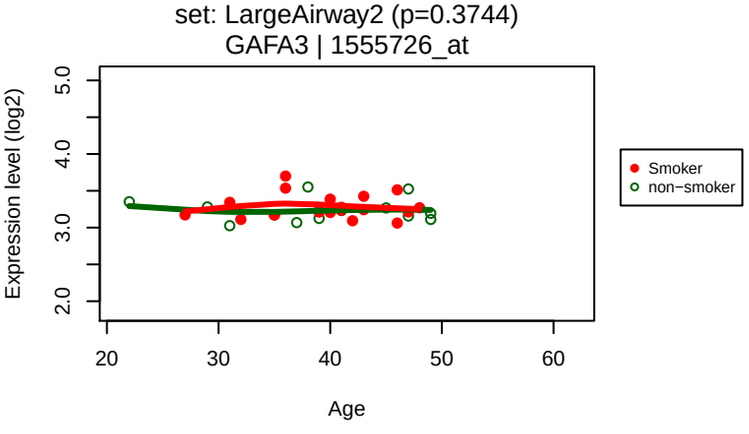

Gene expression ratio (smokers/nonsmokers)
gene: GAFA3 | 1555726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0366
p=0.6831
p=0.0518
p=0.618
p=0.3744
p=0.0087
p=0.1095
p=0.9476
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

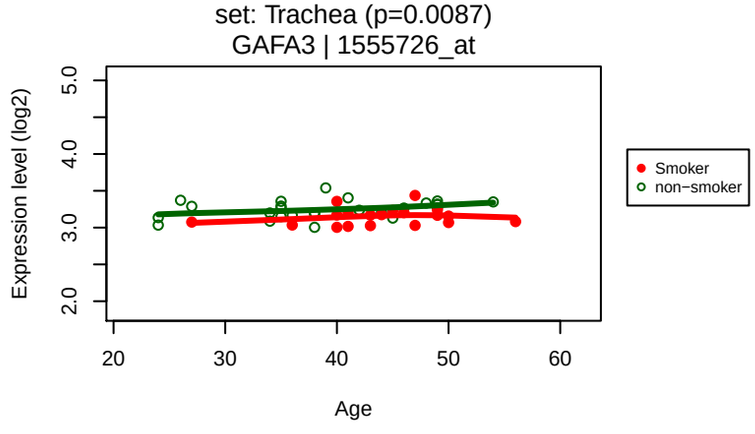

Gene expression ratio in Males
gene: GAFA3 | 1555726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4632
p=0.1967
p=0.0727
p=0.8559
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAFA3 | 1555726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7539
p=0.0356
p=0.0282
p=0.6215
SmallAirway2
LargeAirway2
Trachea
Macrophage2

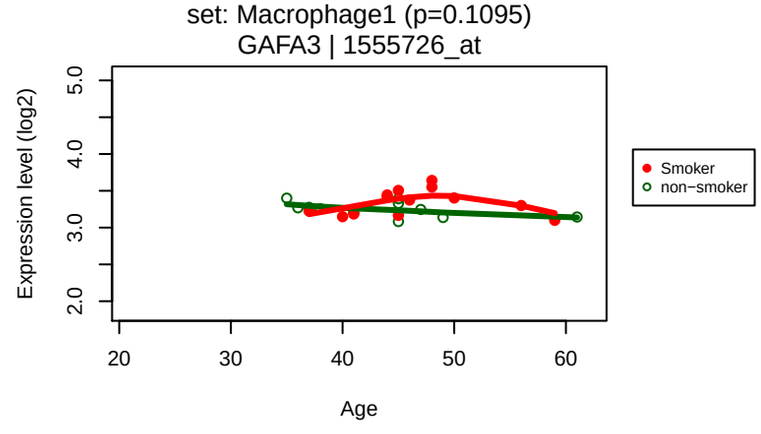

set: SmallAirway1 (p=0.0366)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker

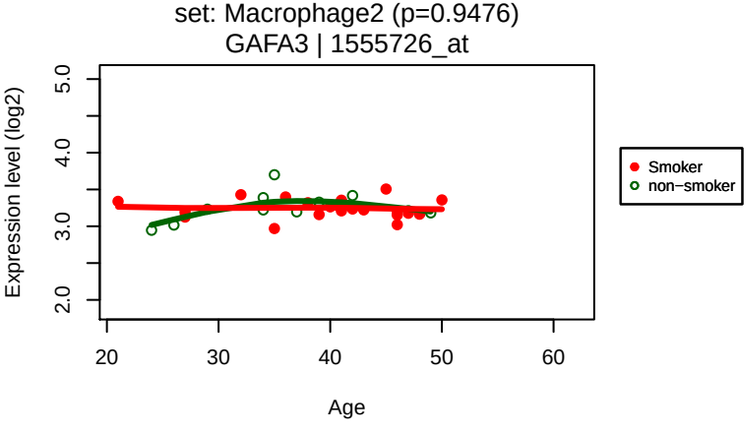

set: SmallAirway2 (p=0.6831)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker

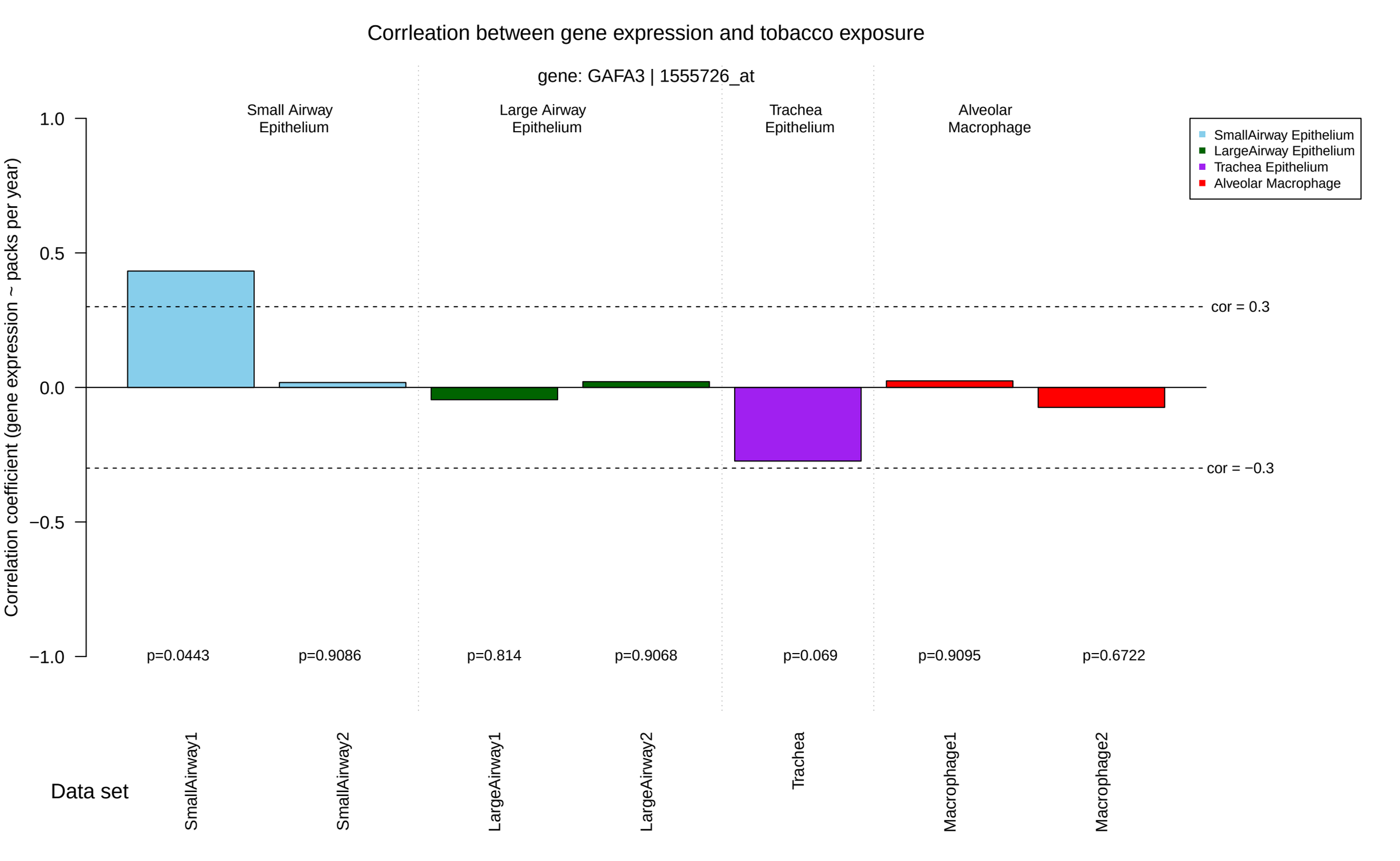

set: LargeAirway1 (p=0.618)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3744)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0087)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1095)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9476)
GAFA3 | 1555726_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAFA3 | 1555726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0443
p=0.9086
p=0.814
p=0.9068
p=0.069
p=0.9095
p=0.6722
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2